Experiment 9: TechSavvy Online Shopping System

Aim

To develop an Online Shopping System using HTML, CSS, and JavaScript that allows users to view products, search and filter items, add products to a shopping cart, calculate the total price, and generate an order confirmation.

Algorithm

1. Start the program and display the online shopping webpage.
2. Display the available products with their names, prices, and categories.
3. Allow users to search for products and filter them by category.
4. Allow users to add products to the shopping cart and increase or decrease quantities.
5. Calculate and display the total price of the selected products.
6. Generate an order ID and display the order confirmation when the user places an order.
7. Allow users to print the order receipt.
8. Stop the program.

Code:

```
<!DOCTYPE html>
<html>
<head>
  <title>TechSavvy Online Shopping</title>
  <style>
    body {
      font-family: Arial;
```

```
    margin: 0;
    background: #f2f5f9;
}

header {
    background: #172554;
    color: white;
    text-align: center;
    padding: 20px;
}

.container {
    width: 90%;
    margin: 20px auto;
}

input, select, button {
    padding: 10px;
    margin: 5px;
}

input, select {
    box-sizing: border-box;
    max-width: 100%;
}
```

```
.products {
    display: flex;
    flex-wrap: wrap;
    gap: 15px;
}

.product {
    background: white;
    padding: 18px;
    width: 220px;
    border-radius: 8px;
    box-shadow: 0 2px 6px #ccc;
    text-align: center;
}

.product span {
    font-size: 45px;
}

button {
    background: #2563eb;
    color: white;
    border: none;
    border-radius: 5px;
    cursor: pointer;
}
```

```
        button:hover {
            background: #1d4ed8;
        }

        .cart {
            background: white;
            padding: 20px;
            margin-top: 25px;
            border-radius: 8px;
        }
    </style>
</head>

<body>

<header>
    <h1>TechSavvy</h1>
    <p>Online Electronics Shopping</p>
</header>

<div class="container">

    <h2>Browse Products</h2>

    <input type="text" id="search"
```

```
        placeholder="Search products"
        oninput="showProducts()">

    <select id="category" onchange="showProducts()">
      <option value="All">All Categories</option>
      <option value="Mobile">Mobiles</option>
      <option value="Laptop">Laptops</option>
      <option value="Accessory">Accessories</option>
    </select>

    <div class="products" id="products"></div>

    <div class="cart">
      <h2>🛒 Shopping Cart</h2>
      <div id="cart"></div>
      <h3 id="total">Total: ₹0</h3>
      <button onclick="checkout()">Place Order</button>
    </div>

    <div id="result"></div>

</div>

<script>
const items = [
    {name: "Smartphone", category: "Mobile", price: 15000, icon: " 📱 "},
```

```
    {name: "Laptop", category: "Laptop", price: 45000, icon: " 💻 "},
    {name: "Headphones", category: "Accessory", price: 1500, icon: " 🎧 "},
    {name: "Smart Watch", category: "Accessory", price: 3000, icon:
" ⌚ "},
    {name: "Tablet", category: "Mobile", price: 12000, icon: " 📲 "}
];

let cart = [];

function showProducts() {
    let search = document.getElementById("search").value.toLowerCase();
    let category = document.getElementById("category").value;

    let list = items.filter(p =>
        p.name.toLowerCase().includes(search) &&
        (category === "All" || p.category === category)
    );

    document.getElementById("products").innerHTML =
        list.map((p, i) => `
            <div class="product">
                <span>${p.icon}</span>
                <h3>${p.name}</h3>
                <p>${p.category}</p>
                <h3>₹${p.price}</h3>
                <button onclick="addCart('${p.name}')">
```

```
                Add to Cart
              </button>
            </div>
        `).join("");

    if (list.length === 0) {
        document.getElementById("products").innerHTML =
            "<p>No products found.</p>";
    }
}

function addCart(name) {
    let product = items.find(p => p.name === name);
    let found = cart.find(p => p.name === name);

    if (found) {
        found.qty++;
    } else {
        cart.push({...product, qty: 1});
    }

    showCart();
}

function showCart() {
    let output = "";
```

```
  let total = 0;

  cart.forEach((p, i) => {
    total += p.price * p.qty;

    output += `
      <p>
        ${p.name} - ₹${p.price} × ${p.qty}
        <button onclick="removeItem(${i})">Remove</button>
      </p>
    `;
  });

  document.getElementById("cart").innerHTML =
    output || "Your cart is empty.";

  document.getElementById("total").innerText =
    "Total: ₹" + total;
}

function removeItem(i) {
  cart.splice(i, 1);
  showCart();
}

function checkout() {
```

```
    if (cart.length === 0) {
        alert("Your cart is empty!");
        return;
    }

    let total = cart.reduce((sum, p) =>
        sum + p.price * p.qty, 0);

    let orderId = "ORD" + Math.floor(10000 + Math.random() * 90000);

    document.getElementById("result").innerHTML = `
        <div class="cart">
            <h2>✅ Order Placed Successfully!</h2>
            <p>Order ID: ${orderId}</p>
            <p>Total Amount: ₹${total}</p>
            <p>Status: Order Placed</p>
            <button onclick="window.print()">Print Receipt</button>
        </div>
    `;

    cart = [];
    showCart();
}

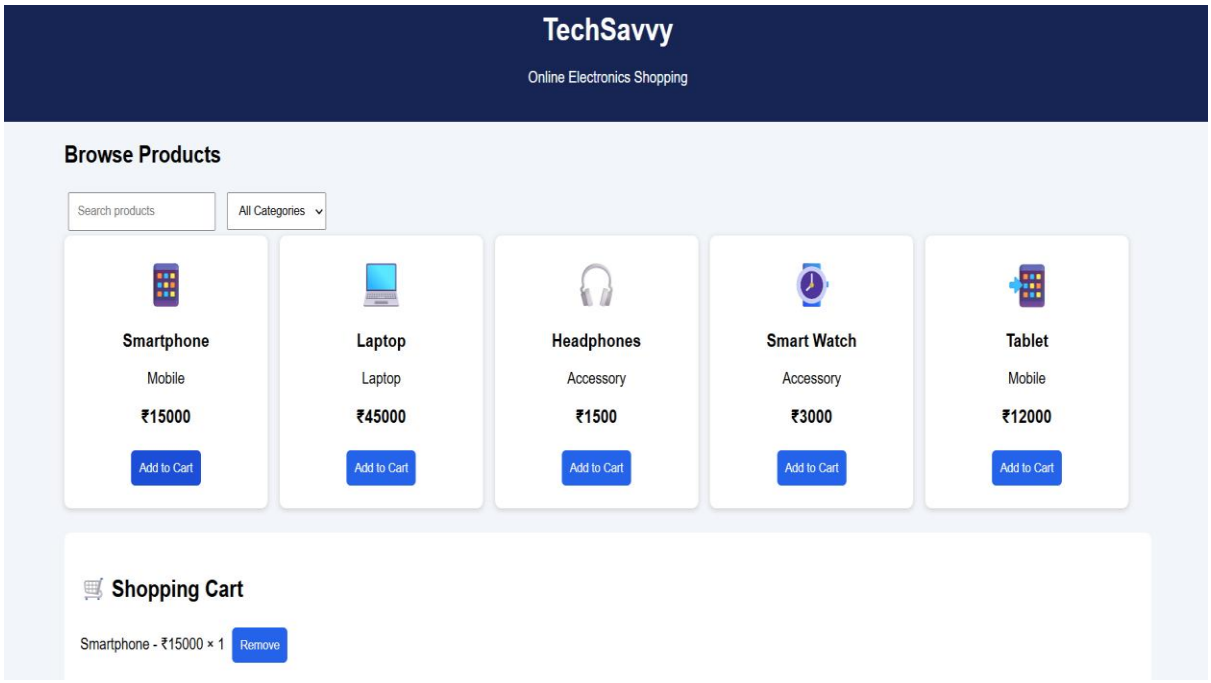
showProducts();
showCart();
```

</script>

</body>

</html>

Output:



Result

Thus, the TechSavvy Online Shopping System was successfully developed using HTML, CSS, and JavaScript. The system supports product searching, category filtering, shopping cart management, total price calculation, and order confirmation.